Code No: 156DR **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, February - 2023
FUNDAMENTALS OF INTERNET OF THINGS
(Common to CE, EEE, ME, CSE, EIE, IT, MCT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

JNTUH Used papers 2023

PART – A

(25 Marks)

1.a)	Define 6LoWPAN.	[2]
b)	Explain about transport layer.	[3]
c)	What is an actuator?	[2]
d)	What are various types of sensors? Explain.	[3]
e)	What are advantages of Raspberry Pi?	[2]
f)	Describe classes in Python programming.	[3]
g)	Define SDN.	[2]
h)	Discuss about data flow in MapReduce.	[3]
i)	What are advantages of Cloud?	[2]
j)	Explain about connected vehicles using IoT.	[3]

PART – B

(50 Marks)

2.a)	Explain about IoT communication APIs in detail.	
b)	What are communication protocols? Explain.	[5+5]
	OR	
3.a)	Discuss about IoT functional blocks.	
b)	What are IoT protocols? Explain.	[5+5]
4.	What is Arduino Programming? Explain about basics of Arduino coding.	[10]
	OR	
5.	Explain in detail about interoperability in IoT.	[10]
6.	Discuss about Python packages of interest for IoT in detail.	[10]
	OR	
7.	Explain about implementation of IoT with Raspberry Pi.	[10]
8.	What are SDN layers? Explain in detail with a neat diagram.	[10]
	OR	
9.a)	Explain about Hadoop cluster setup.	
b)	Discuss about setting up a Strom cluster.	[5+5]
10.	Explain about home automation using IoT in detail.	[10]
	OR	
11.	Discuss about environment monitoring using IoT in detail.	[10]

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